DSP Experiment 1 – Signal Theory Viva

Basic signal concepts, waveform properties, and "what if?" variations

Signals used in the experiment

- Signal 1: $x_1(t) = \cos(t)$, $0 \le t \le 10$
- Signal 2: $x_2(t) = 3\cos(3\pi t + \pi/3)$, $0 \le t \le 8/3$ (four periods)
- Signal 3: $x_3(t) = 2e^{-0.9t}$, $0 \le t \le 10$

Essential concepts

Signal: A quantity that varies with one or more independent variables and carries information.

Continuous-time signal: A signal defined at every instant of time in an interval; written as x(t).

Discrete-time signal: A sequence defined only at distinct time instants; written as x[n].

Analog signal: A physical signal whose value can vary continuously in time and amplitude.

Digital signal: A signal represented using discrete-time samples and quantized amplitude values.

Amplitude: The vertical magnitude or strength of a signal.

Peak amplitude: The maximum magnitude measured from the zero reference level.

Peak-to-peak amplitude: Maximum value minus minimum value. For $A\cos(\omega t)$, it is $2|A|$.

Period (T): The time taken for one complete repetition of a periodic signal. Unit: second.

Frequency (f): The number of cycles per second. Unit: hertz (Hz).

Angular frequency (ω): Rate of phase change in radians per second. $\omega = 2\pi f$.

Phase: The angular position of a sinusoid relative to a reference.

Phase shift: A horizontal displacement of a waveform relative to a reference waveform.

Offset/DC component: A constant value added to a signal, shifting it vertically.

Mean value: The average value of a signal over a specified interval.

RMS value: The square root of the mean of the squared signal; relates to equivalent power.

Periodic signal: A signal satisfying $x(t + T) = x(t)$ for some $T > 0$.

Aperiodic signal: A signal that does not repeat at a fixed interval.

Even signal: A signal satisfying $x(-t) = x(t)$; it is symmetric about the vertical axis.

Odd signal: A signal satisfying $x(-t) = -x(t)$; it has origin symmetry.

Energy signal: A signal with finite total energy and zero average power under the usual infinite-duration definitions.

Power signal: A signal with finite, nonzero average power and typically infinite total energy.

Exponential signal: A signal whose value changes according to an exponential function.

Time constant (τ): For e^{-at}, $\tau = 1/a$; after one time constant, the magnitude falls to 1/e of its initial value.

Decay rate: The positive coefficient a in e^{-at}; a larger a means faster decay.

Growth: An exponential whose magnitude increases with time, such as e^{at} for $a > 0$.

Viva questions and answers

A. Very basic signal questions

Q. What is a signal?

Answer: A time-varying or otherwise varying quantity that conveys information.

Q. What does x(t) mean?

Answer: The value of signal x at time t.

Q. What does the symbol t represent?

Answer: Time, usually measured in seconds.

Q. What is the horizontal axis in a time-domain signal graph?

Answer: Time.

Q. What is the vertical axis?

Answer: Signal amplitude or signal value.

Q. What is the difference between a signal and a system?

Answer: A signal carries information; a system processes or transforms an input signal into an output signal.

Q. What is the difference between continuous-time and discrete-time signals?

Answer: A continuous-time signal exists at every time instant; a discrete-time signal exists only at specified instants.

Q. What is an analog signal?

Answer: A signal that can vary continuously in time and amplitude.

Q. What is a digital signal?

Answer: A signal represented by discrete samples with quantized values.

Q. Can a continuous-time signal have a changing amplitude?

Answer: Yes. Continuous-time describes the time variable, not a requirement that amplitude stay constant.

Q. What is a waveform?

Answer: The shape of a signal when represented graphically.

Q. What is the time domain?

Answer: The representation of a signal as a function of time.

Q. What is the frequency domain?

Answer: A representation showing the frequency components contained in a signal.

Q. What is a periodic signal?

Answer: A signal that repeats its pattern after a fixed interval.

Q. What is a non-periodic signal?

Answer: A signal that does not repeat exactly after any fixed positive interval.

Q. Give examples of periodic signals.

Answer: Ideal sine, cosine, and repeating square waves.

Q. Give examples of non-periodic signals.

Answer: A single pulse, a decaying exponential, or a one-time transient.

Q. What is a cycle?

Answer: One complete repetition of a periodic waveform.

Q. What is one complete oscillation?

Answer: One full cycle, returning to the same phase and value while following the same direction.

Q. What is the unit of amplitude?

Answer: It depends on what the signal represents; for a voltage signal, it is volts.

Q. What is the unit of time?

Answer: Second (s).

Q. What is the unit of frequency?

Answer: Hertz (Hz), equivalent to cycles per second.

Q. What is the unit of angular frequency?

Answer: Radians per second (rad/s).

B. Amplitude, frequency, period, and phase

Q. What happens if the amplitude is doubled?

Answer: The waveform becomes twice as tall in magnitude; frequency and period remain unchanged.

Q. What happens if amplitude is halved?

Answer: The waveform becomes half as tall; its timing and period do not change.

Q. What happens if amplitude is multiplied by –1?

Answer: The waveform is reflected vertically about the time axis.

Q. What happens if 2 is added to a signal?

Answer: The entire waveform shifts upward by 2 units; its shape and period stay the same.

Q. What happens if 2 is subtracted from a signal?

Answer: The waveform shifts downward by 2 units.

Q. What happens if a constant is added to a cosine?

Answer: It creates a vertical offset; the sinusoidal amplitude about its midline is unchanged.

Q. What happens if frequency is doubled?

Answer: The signal completes twice as many cycles per second and its period becomes half.

Q. What happens if frequency is halved?

Answer: The signal completes half as many cycles per second and its period doubles.

Q. What happens if angular frequency is increased?

Answer: The waveform oscillates faster and its period becomes shorter.

Q. What happens if angular frequency is decreased?

Answer: The waveform oscillates more slowly and its period becomes longer.

Q. What happens if the period is doubled?

Answer: The frequency is halved.

Q. What happens if the period is halved?

Answer: The frequency is doubled.

Q. What happens if phase is changed?

Answer: The waveform shifts horizontally; amplitude, frequency, and period remain unchanged.

Q. What happens if phase increases from 0 to $\pi/2$ in $\cos(\omega t + \phi)$?

Answer: The waveform shifts left by one-quarter of a period.

Q. What happens if phase decreases from 0 to $-\pi/2$?

Answer: The waveform shifts right by one-quarter of a period.

Q. What is the phase shift in time for $x(t) = \cos(\omega t + \phi)$?

Answer: The time shift is $-\phi/\omega$; a positive ϕ means a shift to the left.

Q. Does changing phase change the frequency?

Answer: No. It changes only the horizontal position.

Q. Does changing amplitude change frequency?

Answer: No. Amplitude changes vertical scale only.

Q. What is the peak-to-peak value of 4cos(ωt)?

Answer: 8.

Q. What is the peak-to-peak value of −5cos(ωt)?

Answer: 10; its maximum is 5 and minimum is −5.

Q. What is the maximum and minimum of A cos(ωt) when A is positive?

Answer: Maximum A and minimum −A.

Q. What is the maximum and minimum of A cos(ωt) + C?

Answer: C + |A| and C − |A|.

Q. What does a negative frequency mean in a real cosine expression?

Answer: A cosine with negative angular frequency traces the same real waveform as the corresponding positive frequency, with phase interpreted accordingly.

Q. What is the relation between f and T?

Answer: f = 1/T and T = 1/f.

Q. What is the relation between ω and f?

Answer: ω = 2πf.

Q. How many radians are in one complete cycle?

Answer: 2π radians, or 360 degrees.

Q. How many degrees are in half a cycle?

Answer: 180 degrees.

Q. How many degrees are in a quarter cycle?

Answer: 90 degrees.

C. Questions about the three experiment signals

Q. What type of signal is cos(t)?

Answer: A cosine wave, which is periodic and continuous-time.

Q. What are the amplitude, angular frequency, and phase of cos(t)?

Answer: Amplitude 1, angular frequency 1 rad/s, phase 0.

Q. What is the period of cos(t)?

Answer: 2π seconds.

Q. What is the frequency of cos(t)?

Answer: 1/(2π) Hz, approximately 0.1592 Hz.

Q. What is cos(0)?

Answer: 1.

Q. What is the value of cos(t) at t = π/2?

Answer: 0.

Q. What is the value of cos(t) at t = π?

Answer: −1.

Q. What is the value of cos(t) at t = 2π?

Answer: 1; one complete cycle has elapsed.

Q. What is the amplitude of 3cos(3πt + π/3)?

Answer: 3.

Q. What is its angular frequency?

Answer: 3π rad/s.

Q. What is its frequency?

Answer: 1.5 Hz.

Q. What is its period?

Answer: 2/3 second.

Q. What is its phase?

Answer: π/3 radians, or 60 degrees.

Q. Why does the second cosine start at 1.5?

Answer: At t = 0, it equals 3cos(π/3) = 3 × 1/2 = 1.5.

Q. What are the maximum and minimum values of the second cosine?

Answer: Maximum 3 and minimum −3.

Q. How many cycles occur in 8/3 seconds?

Answer: Four cycles, because (8/3)/(2/3) = 4.

Q. What type of signal is 2e^(−0.9t)?

Answer: A decaying exponential signal.

Q. What is its initial value?

Answer: 2.

Q. What is its value as t becomes very large?

Answer: It approaches zero.

Q. Is the decaying exponential periodic?

Answer: No.

Q. What is its time constant?

Answer: 1/0.9 ≈ 1.111 seconds.

Q. What happens if the 2 in 2e^(−0.9t) becomes 4?

Answer: The initial value doubles to 4 and the whole curve doubles vertically; the decay rate and time constant stay the same.

Q. What happens if 0.9 becomes 1.8?

Answer: It decays twice as fast; the time constant halves.

Q. What happens if 0.9 becomes 0.45?

Answer: It decays more slowly; the time constant doubles.

Q. What happens if the negative exponent becomes positive?

Answer: The signal grows exponentially instead of decaying.

Q. What happens if a constant 3 is added to the exponential?

Answer: The curve shifts upward by 3 and approaches 3 instead of zero.

D. "What if?" waveform modifications

Q. For x(t) = A cos(ωt + ϕ), what if A changes from 1 to 5?

Answer: Peak amplitude becomes 5 and peak-to-peak becomes 10; period and phase do not change.

Q. What if x(t) = 2cos(t) becomes 2cos(2t)?

Answer: Amplitude stays 2; angular frequency doubles and period halves from 2π to π.

Q. What if x(t) = cos(t) becomes cos(0.5t)?

Answer: Amplitude stays 1; angular frequency halves and period doubles to 4π.

Q. What if cos(t) becomes sin(t)?

Answer: The waveform is shifted in phase by 90 degrees relative to cosine: $\sin(t) = \cos(t - \pi/2)$.

Q. What if x(t) = cos(t) becomes cos(t + π)?

Answer: The signal is inverted: $\cos(t + \pi) = -\cos(t)$.

Q. What if x(t) = cos(t) becomes cos(t − π)?

Answer: It is also inverted, because a shift of $\pm\pi$ gives $-\cos(t)$.

Q. What if x(t) = cos(t) becomes cos(t + 2π)?

Answer: There is no change; adding 2π to phase gives the same waveform.

Q. What if x(t) = cos(t) becomes 3cos(t) + 2?

Answer: Amplitude becomes 3 and the waveform is shifted upward by 2; maximum 5, minimum −1.

Q. What if x(t) = 3cos(t) − 2?

Answer: Amplitude is 3 with a downward offset of 2; maximum 1 and minimum −5.

Q. What if the signal is multiplied by 0.5?

Answer: All amplitudes are halved; the waveform timing is unchanged.

Q. What if the signal is multiplied by −2?

Answer: It is vertically inverted and its amplitude magnitude doubles.

Q. What if x(t) is replaced by x(2t)?

Answer: Time is compressed by a factor of 2; the signal runs twice as fast and its period halves.

Q. What if x(t) is replaced by x(t/2)?

Answer: Time is expanded by a factor of 2; the signal runs half as fast and its period doubles.

Q. What if x(t) is replaced by x(t − 3)?

Answer: The entire signal is delayed by 3 seconds, shifting it to the right.

Q. What if x(t) is replaced by x(t + 3)?

Answer: The signal is advanced by 3 seconds, shifting it to the left.

Q. What if x(t) is replaced by x(−t)?

Answer: The signal is time-reversed, reflected about $t = 0$.

Q. What if a high-frequency sinusoid is added to a low-frequency sinusoid?

Answer: The result is a sum waveform containing both frequency components; it may look more complex.

Q. What if two identical in-phase sinusoids are added?

Answer: Their amplitudes add, doubling the amplitude while frequency and phase remain the same.

Q. What if two equal sinusoids 180 degrees out of phase are added?

Answer: They cancel completely, producing zero.

Q. What if two sinusoids have the same frequency but different phases?

Answer: They combine into another sinusoid of that same frequency, with a resultant amplitude and phase.

Q. What if two sinusoids have different frequencies?

Answer: Their sum generally contains both frequencies and is not a single pure sinusoid.

Q. What if random noise is added to a signal?

Answer: The waveform becomes irregular; the original signal may become harder to distinguish.

Q. What if a signal is clipped above a fixed limit?

Answer: Values above the limit are flattened at that limit, distorting the waveform.

Q. What if a signal is rectified using absolute value?

Answer: Negative portions are flipped positive; the output is nonnegative.

Q. What if a signal is squared?

Answer: Negative values become positive; the waveform shape and frequency content change.

E. Signal properties and classifications

Q. Is cosine an even or odd signal?

Answer: Cosine is even: cos(−t) = cos(t).

Q. Is sine an even or odd signal?

Answer: Sine is odd: sin(−t) = −sin(t).

Q. Is a constant signal periodic?

Answer: Yes; any positive period satisfies the periodicity condition, though it has no unique fundamental period.

Q. Is a decaying exponential periodic?

Answer: No, because it does not repeat.

Q. Is a sinusoid always periodic?

Answer: A continuous-time sinusoid with nonzero fixed frequency is periodic.

Q. What is a bounded signal?

Answer: A signal whose magnitude never exceeds some finite value.

Q. Is a cosine signal bounded?

Answer: Yes; its values remain between −|A| and |A| if there is no offset.

Q. Is 2e^(−0.9t) bounded for t ≥ 0?

Answer: Yes; it remains between 0 and 2.

Q. What is a causal signal?

Answer: A signal that is zero for all t < 0.

Q. What is a right-sided signal?

Answer: A signal that is zero before some time and exists to the right of that time.

Q. What is a left-sided signal?

Answer: A signal that exists to the left of a time boundary and is zero afterward.

Q. What is a finite-duration signal?

Answer: A signal that is nonzero only over a finite time interval.

Q. What is a unit step signal?

Answer: u(t) = 0 for t < 0 and u(t) = 1 for t > 0; its value at t = 0 depends on convention.

Q. What is a unit impulse?

Answer: An idealized impulse with zero duration and unit area, used to model instantaneous events.

Q. What is a ramp signal?

Answer: A signal that increases linearly with time, commonly r(t) = t u(t).

Q. What is a DC signal?

Answer: A constant signal that does not vary with time.

Q. What is an AC signal?

Answer: A signal that varies with time, often alternating around a reference level.

Q. Can a signal contain both AC and DC components?

Answer: Yes; for example, A cos(ωt) + C contains a sinusoidal AC component and a DC offset C.

F. Slight changes to exponential and sinusoidal signals

Q. In Ae^(−at), what does A control?

Answer: Initial amplitude, since x(0) = A.

Q. In Ae^(−at), what does a control?

Answer: The speed of decay; larger positive a means faster decay.

Q. What is the time constant for Ae^(−at)?

Answer: τ = 1/a, for a > 0.

Q. After one time constant, what fraction remains?

Answer: 1/e ≈ 36.8% of the initial value.

Q. After two time constants, approximately what fraction remains?

Answer: e^(−2) ≈ 13.5%.

Q. After three time constants, approximately what fraction remains?

Answer: e^(−3) ≈ 5.0%.

Q. What if A is negative in Ae^(−at)?

Answer: The signal starts negative and approaches zero from below for a > 0.

Q. What if a = 0?

Answer: The exponential becomes constant: Ae^0 = A.

Q. What if a is negative in Ae^(−at)?

Answer: The exponent becomes positive in time, so the magnitude grows exponentially.

Q. What if the exponential is e^(−at) and a is doubled?

Answer: The time constant is halved and the decay is faster.

Q. In A cos(ωt + ϕ) + C, what controls the midline?

Answer: C; the waveform oscillates around y = C.

Q. In A cos(ωt + ϕ) + C, what controls the height from midline to peak?

Answer: |A|.

Q. In A cos(ωt + ϕ), what controls how tightly packed the cycles are?

Answer: ω; larger angular frequency gives more closely spaced cycles.

Q. If the amplitude doubles but the frequency stays the same, do zero crossings move?

Answer: No, assuming no offset or phase change; the same-time zero crossings remain.

Q. If a DC offset is added, do the peaks and troughs move?

Answer: Yes, both shift vertically by the offset, but the peak-to-peak value stays the same.

Q. If phase changes, do maximum and minimum values change?

Answer: No, not for an unchanged amplitude and offset.

Q. If frequency changes, do maximum and minimum values change?

Answer: No, not for an unchanged amplitude and offset; only their timing changes.

Quick formulas to remember

- Sinusoid: x(t) = A cos(ωt + ϕ) + C
- Frequency: f = ω/(2π)
- Period: T = 1/f = 2π/ω
- Peak-to-peak value: 2|A|
- Time shift in cos(ωt + ϕ): −ϕ/ω
- Time scaling: x(kt), k > 1 compresses time and multiplies frequency by k
- Exponential decay: x(t) = Ae^(−at), a > 0

- Exponential time constant: $\tau = 1/a$

Important examiner traps

- Amplitude changes vertical size; frequency changes how quickly cycles repeat; phase changes horizontal alignment.
- Adding a constant shifts the waveform vertically. It does not change peak-to-peak amplitude.
- Multiplying the input time by 2 compresses the waveform; replacing t by t/2 stretches it.
- For $\cos(\omega t + \phi)$, a positive phase term shifts the graph left, not right.
- The second experiment sinusoid has frequency 1.5 Hz but period 2/3 second.
- A decaying exponential approaches zero; it is not a periodic waveform.
- The time constant is inversely proportional to the decay rate.